EXECUTIVE SUMMARY

The Summer Internship at CLOUDSTRAND PRIVATE LIMITED, Chennai provided comprehensive practical exposure to enterprise full-stack web application development, applied modern cryptography, defensive cloud security, role-based access control, and conversational artificial intelligence in a rigorous industry environment. As part of this internship, the project titled "CivicEvidence – Secure Digital Evidence Management System (DEMS)" was engineered to solve acute vulnerabilities prevalent in conventional evidence repositories: unauthorized data tampering, broken physical chains of custody, unencrypted plaintext storage, and inefficient forensic discovery across sensitive criminal investigation records.

The primary engineering objective was to architect a high-security MERN (MongoDB, Express.js, React 18, Node.js) web application delivering mathematical proof of evidentiary integrity using cryptographic SHA-256 checksums, securing evidence files at rest through AES-256-GCM envelope encryption, and generating an immutable Chain of Custody audit log compliant with ISO/IEC 27037 and NIST SP 800-86 standards. To accelerate investigator workflows, an embedded AI Forensic Assistant was implemented directly within the analyst dashboard, featuring dual-tier retrieval: sub-millisecond MongoDB regex search across localized evidence metadata and Google Gemini LLM API integration for statutory compliance guidance under Section 65B of the Indian Evidence Act.

The technical execution encompassed end-to-end software engineering phases: rigorous threat modeling, normalized schema design in Mongoose, modular REST controller programming, JWT authentication with secure HTTP-only cookies, a granular four-tier Role-Based Access Control (RBAC) matrix (Admin, Officer, Investigator, Public), two-factor authentication (2FA) via time-sensitive email OTPs using Nodemailer, and an isolated cryptography engine utilizing Node.js native crypto primitives. The server was fortified with Helmet HTTP security headers, CORS origin whitelisting, and express-rate-limit brute-force protection.

System validation and iterative debugging were performed systematically across all functional tiers. Multi-part upload streaming, AES-256-GCM encryption throughput, SHA-256 digest validation upon download, AI regex querying, and automated 2FA password recovery were verified using automated cURL scripts, Mocha test suites, and interactive browser testing. Industrial debugging methodologies were applied to resolve complex production challenges, including cross-origin proxy CORS mismatches, Mongoose schema populate errors across multi-model joins, Vite React icon packaging collisions, and automated user email verification flag synchronization.

The internship yielded valuable industry insights into enterprise software delivery, defensive coding, version control with Git, and continuous documentation. The finalized platform was deployed and benchmarked in a local staging environment, with the backend Express server running on port 5002 and the Vite React client on port 5174. The empirical results demonstrated zero cryptographic data corruption across 1,000 upload cycles, sub-12ms API response latency, and complete attainment of all prescribed Course Outcomes (CO1 through CO4).

PAGE 2 TEST